Deneb

MYMENSINGH GIRLS' CADET COLLEGE
PROGRESS TEST EXAMINATION - 2025
CLASS: HSC
MULTIPLE CHOICE QUESTIONS
STATISTICS
SECOND PAPER
[According to the Syllabus of 2026]
TIME – 20 minutes
FULL MARKS – 20

Subject Code: | 1 | **3** | 0 |

Set: **Ga**

[N.B. – Answer all the questions. Each question carries ONE mark. Block fully, with a black ball- point pen, the circle of the letter that stands for the correct/best answer in the "Answer sheet" for the Multiple Choice Questions Examination.]

Candidates are asked not to leave any mark or spot on the question paper.

1. **How many samples of size 4 can be drawn without replacement from a population of size 10?**

   (a) 200 (b) 220 (c) 210 (d) 550

2. **In sampling with replacement, the probability of each selected sample is –**

   (a) $\frac{1}{N^n}$ (b) $\frac{1}{{}^{N}C_n}$ (c) $\frac{1}{N \times n}$ (d) $\frac{n}{N}$

3. **Which of the following are true about stratified sampling?**

   i. The population is divided into homogeneous subgroups called strata
   ii. Samples are drawn independently from each stratum
   iii. The strata are heterogenous

   Which one is correct?

   (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

4. **In a survey of 60 people, 35 liked coffee, 28 liked tea, and 12 liked neither. If a person is selected at random, what is the probability that they like at least one of the two beverages?**

   (a) $\frac{3}{5}$ (b) $\frac{4}{5}$ (c) $\frac{2}{3}$ (d) $\frac{5}{6}$

Answer the next three questions based on the following information

X	1	2	3
P(x)	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{1}{2}$

5. **What is $P(X \geq 2)$?**

   (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{2}{3}$ (d) $\frac{5}{6}$

6. **Find the cumulative probability $F(2)$.**

   (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{1}{6}$ (d) $\frac{2}{3}$

7. $P(X = 1$ **or** $X = 3)$ **equals**

   (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{5}{6}$ (d) $\frac{2}{3}$

8. $P(A \cup B) = P(A) + P(B)$ **implies A & B are –**

   (a) Disjoint (b) Independent (c) Joint (d) Independent

Answer the next THREE questions based on the following information.

A card is drawn at random from a standard deck of 52 playing cards.

9. **What is the probability that the card is a Queen?**

   (a) 0.0192 (b) 0.0769 (c) 0.25 (d) 0.5

10. **P(The card is not a Spade)**

    (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{3}{4}$ (d) $\frac{1}{3}$

11. **P(The card is black or a Heart)**

    (a) $\frac{1}{2}$ (b) $\frac{3}{4}$ (c) $\frac{1}{3}$ (d) $\frac{1}{4}$

12. $P(A)=\frac{2}{3}$, $P(B)=\frac{1}{5}$, **and** $P(A\cap B)=\frac{1}{6}$

i. A and B are independent
ii. $P(A\cup B)=\frac{7}{10}$
iii. $P(\bar{A})=\frac{1}{3}$

Which one is correct?

(a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

13. **Which reflectes an unchanging population?**

(a) GRR=1 (b) NRR>1 (c) GRR<1 (d) GRR=NRR

14. **A bacterial culture grows exponentially with a growth rate constant of $r=0.2$ per hour. How many hours will it take to triple?**

(a) $\frac{\ln 3}{0.2}$ (b) $\frac{\ln 2}{0.2}$ (c) $0.2\times\ln 3$ (d) $0.2\times\ln 2$

15. **Which one is correct?**

(a) TFR>GRR>NRR (b) TFR=GRR=NRR (c) TFR<GRR<NRR (d) NRR>GRR>TFR

16. **If $V(X)=2$, what is the variance of $(2X+m)$, where m is a constant?**

(a) 0 (b) 2 (c) 4 (d) 8

Answer the next three questions based on the following information

The probability function of random variable x is given below:

$$P(x)=\frac{2x+1}{k};x=1,2,3,4$$

17. **What is the value of k?**

(a) 18 (b) 25 (c) 12 (d) 24

18. **What is $E(X)$?**

(a) 1.75 (b) 2.92 (c) 3.25 (d) 2.25

19. **What is $V(X)$?**

(a) 1.05 (b) 3.0 (c) 1.5 (d) 1.25

20. **For a discrete random variable X with probability mass function $p(x)$:**

i. $P(a\le X\le b)=\sum_{x=a}^{b}p(x)$
ii. $\sum_{x}p(x)=1$
iii. $p(x)\ge 0$ for all x

Which one is correct?

(a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii